

Asssignment 5 - Answer Sheet

Sheet: All Errors

File	Error Category	Error Type	Line	Code Snippet	Fix
1.c	Syntax	Missing semicolon	6	int marks[5] = {80,90,70,60,50}	Add ; at end of declaration
1.c	Syntax	Missing semicolon	88	return 0	Add ;
1.c	Array Bounds	Out of bounds access	10	for(i=0;i<=5;i++) total += marks[i]	Change to i<5 (marks[5] invalid)
1.c	Array Bounds	Out of bounds assignment	20	scores[5] = 60;	scores size is 5, index 5 invalid
1.c	Array Bounds	Out of bounds access	36	for(i=0;i<=5;i++) printf attendance[i]	Change to i<5
1.c	Type Mismatch	Integer division	14	average = total / 5;	Use total / 5.0
1.c	Type Mismatch	Wrong format specifier	16	printf("Average = %d", average)	Use %f for float
1.c	Buffer Overflow	strcpy overflow	28	strcpy(subject,"Programming")	subject[5] too small, need [12]
1.c	Semantic	Invalid array index	32	attendance["one"] = 1;	Use integer index: attendance[1]
1.c	Semantic	Invalid array index	33	attendance[2.5] = 1;	Use integer index: attendance[2]
1.c	Semantic	Missing & in scanf	45	scanf("%d", arr[i]);	Use &arr[i]
1.c	Type Mismatch	Char vs string	55	grade[0] = "A";	Use single quotes: 'A'
1.c	Semantic	Array assignment	63	inventory = prices;	Use memcpy or loop copy
1.c	Type Mismatch	Wrong format specifier	70	printf("%d", salary[0])	Use %f for float
1.c	Semantic	Array assignment	76	city = "Jaipur";	Use strcpy(city, "Jaipur")
1.c	Array Bounds	Out of bounds access	80	for(i=0;i<=5;i++) result[i]	Change to i<5
2.c	Syntax	Missing semicolon	82	printf("Program Ended ")	Add ;
2.c	Logical	Wrong accumulation	16	sum = numbers[i];	Use sum += numbers[i]
2.c	Logical	Wrong comparison (min/max)	28	if(marks[i] < highest)	Use > to find highest
2.c	Array Bounds	Out of bounds access	10	for(i=0;i<=10;i++) numbers[i]	Change to i<10
2.c	Array Bounds	Out of bounds assignment	40	attendance[5] = 1;	attendance size is 5
2.c	Array Bounds	Out of bounds access	41	for(i=0;i<=5;i++) attendance[i]	Change to i<5
2.c	Array Bounds	Out of bounds access	74	for(i=0;i<=5;i++) inventory[i]	Change to i<5
2.c	Type Mismatch	Integer division	18	average = sum / 10;	Use sum / 10.0
2.c	Type Mismatch	Wrong format specifier	20	printf("Average = %d", average)	Use %f
2.c	Buffer Overflow	strcpy overflow	48	strcpy(subject,"ComputerScience")	subject[10] too small, need [16]
2.c	Semantic	Array assignment	52	branch = "IT";	Use strcpy(branch, "IT")
2.c	Semantic	Invalid array index	58	fees["two"] = 50000;	Use integer index
2.c	Semantic	Invalid array index	59	fees[3.5] = 60000;	Use integer index
2.c	Semantic	Missing & in scanf	68	scanf("%d", result[i]);	Use &result[i]
2.c	Logical	Assignment vs comparison	78	if(stock[2] = 300)	Use ==
2.c	Semantic	Array assignment	82	backup = stock;	Use memcpy or loop copy
2.c	Type Mismatch	Wrong format specifier	88	printf("%d", salary[0])	Use %f

File	Error Category	Error Type	Line	Code Snippet	Fix
2.c	Buffer Overflow	strcpy overflow	96	strcpy(city,"JaipurCity")	city[8] too small, need [11]
2.c	Semantic	Array assignment	102	count = inventory;	Use memcpy or loop copy
2.c	String	Missing null terminator	54	college[0..4] set manually	Add college[5] = '\0'
3.c	Syntax	Missing semicolon	118	return 0	Add ;
3.c	Logical	Integer division	18	average = total / 5;	Use total / 5.0
3.c	Logical	Wrong comparison (min/max)	32	if(marks[i] > lowest)	Use < to find lowest
3.c	Array Bounds	Out of bounds access	42	for(i=0;i<=5;i++) attendance[i]	Change to i<5
3.c	Array Bounds	Out of bounds assignment	56	fees[5] = 100000;	fees size is 5
3.c	Array Bounds	Out of bounds access	96	for(i=0;i<=5;i++) inventory[i]	Change to i<5
3.c	Array Bounds	Out of bounds access	114	for(i=0;i<=5;i++) numbers[i]	Change to i<5
3.c	Buffer Overflow	strcpy overflow	48	strcpy(subject,"Programming")	subject[8] too small, need [12]
3.c	Logical	Assignment vs comparison	60	if(stock[2] = 30)	Use ==
3.c	Semantic	Array assignment	66	backup = stock;	Use memcpy or loop copy
3.c	Type Mismatch	Wrong format specifier	72	printf("%d", salary[0])	Use %f
3.c	Semantic	Missing & in scanf	80	scanf("%d", result[i]);	Use &result[i]
3.c	Semantic	Array assignment	86	city = "Jaipur";	Use strcpy
3.c	Semantic	Invalid array index	92	inventory["one"] = 100;	Use integer index
3.c	Semantic	Invalid array index	93	inventory[2.5] = 200;	Use integer index
3.c	Semantic	Array assignment	106	count = inventory;	Use memcpy or loop copy
3.c	Buffer Overflow	strcpy overflow	112	strcpy(course,"Computer")	course[6] too small, need [9]
3.c	String	Missing null terminator	100	college[0..4] set manually	Add college[5] = '\0'
4.c	Syntax	Missing semicolon	122	printf("Report Generated Successfully ")	Add ;
4.c	Logical	Integer division	14	averageSales = totalSales / 5;	Use totalSales / 5.0
4.c	Logical	Wrong comparison (min/max)	26	if(sales[i] < highestSale)	Use > to find highest
4.c	Array Bounds	Out of bounds assignment	36	monthlySales[5] = 250;	monthlySales size is 5
4.c	Array Bounds	Out of bounds access	38	for(i=0;i<=5;i++) monthlySales[i]	Change to i<5
4.c	Array Bounds	Out of bounds access	78	for(i=0;i<=5;i++) employees[i]	Change to i<5
4.c	Type Mismatch	Wrong format specifier	16	printf("Average Sales = %d", averageSales)	Use %f
4.c	Buffer Overflow	strcpy overflow	44	strcpy(product,"LaptopComputer")	product[8] too small, need [15]
4.c	Semantic	Array assignment	48	category = "Electronics";	Use strcpy
4.c	Semantic	Invalid array index	54	stock["first"] = 100;	Use integer index
4.c	Semantic	Invalid array index	55	stock[2.7] = 200;	Use integer index
4.c	Semantic	Missing & in scanf	64	scanf("%d", orders[i]);	Use &orders[i]
4.c	Logical	Assignment vs comparison	74	if(revenue[2] = 3000)	Use ==
4.c	Semantic	Array assignment	80	backupRevenue = revenue;	Use memcpy or loop copy
4.c	Type Mismatch	Wrong format specifier	86	printf("%d", commission[0])	Use %f

File	Error Category	Error Type	Line	Code Snippet	Fix
4.c	Semantic	Array assignment	90	city = "Jaipur";	Use strcpy
4.c	Buffer Overflow	strcpy overflow	106	strcpy(department,"Marketing")	department[6] too small, need [10]
4.c	Semantic	Array assignment	112	expenses = sales;	Use memcpy or loop copy
4.c	Array Bounds	Negative index	116	ratings[-1] = 5;	Use valid index 0-4
4.c	Type Mismatch	Wrong format specifier	124	printf("%s", feedback[0])	feedback is int[], use %d
4.c	String	Missing null terminator	94	branch[0..3] set manually	Add branch[4] = '\0'
5.c	Logical	Integer division	18	average = total / 10;	Use total / 10.0
5.c	Logical	Wrong comparison (min/max)	28	if(marks[i] < highest)	Use > to find highest
5.c	Logical	Wrong comparison (min/max)	40	if(marks[i] > lowest)	Use < to find lowest
5.c	Array Bounds	Out of bounds assignment	52	attendance[10] = 0;	attendance size is 10 (indices 0-9)
5.c	Array Bounds	Out of bounds access	56	for(i=0;i<=10;i++) attendance[i]	Change to i<10
5.c	Array Bounds	Out of bounds access	130	for(i=0;i<=5;i++) stock[i]	Change to i<5
5.c	Array Bounds	Negative index	154	ratings[-1] = 10;	Use valid index 0-4
5.c	Array Bounds	Out of bounds access	170	printf temp[5]	temp size is 5, index 5 invalid
5.c	Type Mismatch	Wrong format specifier	20	printf("Average = %d", average)	Use %f
5.c	Buffer Overflow	strcpy overflow	64	strcpy(subject,"DataStructures")	subject[10] too small, need [15]
5.c	Semantic	Array assignment	68	department = "InformationTechnology";	Use strcpy
5.c	Semantic	Invalid array index	74	fees["two"] = 50000;	Use integer index
5.c	Semantic	Invalid array index	75	fees[4.5] = 70000;	Use integer index
5.c	Semantic	Missing & in scanf	84	scanf("%d", result[i]);	Use &result[i]
5.c	Logical	Assignment vs comparison	94	if(inventory[2] = 30)	Use ==
5.c	Semantic	Array assignment	100	backup = inventory;	Use memcpy or loop copy
5.c	Type Mismatch	Wrong format specifier	106	printf("%d", salary[0])	Use %f
5.c	Semantic	Array assignment	110	city = "Jaipur";	Use strcpy
5.c	Buffer Overflow	strcpy overflow	138	strcpy(course,"Programming")	course[8] too small, need [12]
5.c	Type Mismatch	Wrong format specifier	160	printf("%s", feedback[0])	feedback is int[], use %d
5.c	Semantic	Array assignment	166	scores = marks;	Use loop to copy first 5 elements
5.c	String	Missing null terminator	116	branch[0..3] set manually	Add branch[4] = '\0'

Sheet: Summary by Category

Error Category	1.c	2.c	3.c	4.c	5.c
Array Bounds	4	4	4	4	5
Buffer Overflow	1	2	2	2	2
Logical	0	3	3	3	4
Semantic	5	6	6	7	7
String	0	1	1	1	1

Error Category	1.c	2.c	3.c	4.c	5.c
Syntax	2	1	1	1	0
Type Mismatch	4	3	1	3	3

Sheet: Total by File

File	Total Errors
1.c	16
2.c	20
3.c	18
4.c	21
5.c	22

Sheet: File 1 Errors

File	Error Category	Error Type	Line	Code Snippet	Fix
1.c	Syntax	Missing semicolon	6	int marks[5] = {80,90,70,60,50}	Add ; at end of declaration
1.c	Syntax	Missing semicolon	88	return 0	Add ;
1.c	Array Bounds	Out of bounds access	10	for(i=0;i<=5;i++) total += marks[i]	Change to i<5 (marks[5] invalid)
1.c	Array Bounds	Out of bounds assignment	20	scores[5] = 60;	scores size is 5, index 5 invalid
1.c	Array Bounds	Out of bounds access	36	for(i=0;i<=5;i++) printf attendance[i]	Change to i<5
1.c	Type Mismatch	Integer division	14	average = total / 5;	Use total / 5.0
1.c	Type Mismatch	Wrong format specifier	16	printf("Average = %d", average)	Use %f for float
1.c	Buffer Overflow	strcpy overflow	28	strcpy(subject,"Programming")	subject[5] too small, need [12]
1.c	Semantic	Invalid array index	32	attendance["one"] = 1;	Use integer index: attendance[1]
1.c	Semantic	Invalid array index	33	attendance[2.5] = 1;	Use integer index: attendance[2]
1.c	Semantic	Missing & in scanf	45	scanf("%d", arr[i]);	Use &arr[i]
1.c	Type Mismatch	Char vs string	55	grade[0] = "A";	Use single quotes: 'A'
1.c	Semantic	Array assignment	63	inventory = prices;	Use memcpy or loop copy
1.c	Type Mismatch	Wrong format specifier	70	printf("%d", salary[0])	Use %f for float
1.c	Semantic	Array assignment	76	city = "Jaipur";	Use strcpy(city, "Jaipur")
1.c	Array Bounds	Out of bounds access	80	for(i=0;i<=5;i++) result[i]	Change to i<5

Sheet: File 2 Errors

File	Error Category	Error Type	Line	Code Snippet	Fix
2.c	Syntax	Missing semicolon	82	printf("Program Ended ")	Add ;
2.c	Logical	Wrong accumulation	16	sum = numbers[i];	Use sum += numbers[i]
2.c	Logical	Wrong comparison (min/max)	28	if(marks[i] < highest)	Use > to find highest

File	Error Category	Error Type	Line	Code Snippet	Fix
2.c	Array Bounds	Out of bounds access	10	for(i=0;i<=10;i++) numbers[i]	Change to i<10
2.c	Array Bounds	Out of bounds assignment	40	attendance[5] = 1;	attendance size is 5
2.c	Array Bounds	Out of bounds access	41	for(i=0;i<=5;i++) attendance[i]	Change to i<5
2.c	Array Bounds	Out of bounds access	74	for(i=0;i<=5;i++) inventory[i]	Change to i<5
2.c	Type Mismatch	Integer division	18	average = sum / 10;	Use sum / 10.0
2.c	Type Mismatch	Wrong format specifier	20	printf("Average = %d", average)	Use %f
2.c	Buffer Overflow	strcpy overflow	48	strcpy(subject,"ComputerScience")	subject[10] too small, need [16]
2.c	Semantic	Array assignment	52	branch = "IT";	Use strcpy(branch, "IT")
2.c	Semantic	Invalid array index	58	fees["two"] = 50000;	Use integer index
2.c	Semantic	Invalid array index	59	fees[3.5] = 60000;	Use integer index
2.c	Semantic	Missing & in scanf	68	scanf("%d", result[i]);	Use &result[i]
2.c	Logical	Assignment vs comparison	78	if(stock[2] = 300)	Use ==
2.c	Semantic	Array assignment	82	backup = stock;	Use memcpy or loop copy
2.c	Type Mismatch	Wrong format specifier	88	printf("%d", salary[0])	Use %f
2.c	Buffer Overflow	strcpy overflow	96	strcpy(city,"JaipurCity")	city[8] too small, need [11]
2.c	Semantic	Array assignment	102	count = inventory;	Use memcpy or loop copy
2.c	String	Missing null terminator	54	college[0..4] set manually	Add college[5] = '\0'

Sheet: File 3 Errors

File	Error Category	Error Type	Line	Code Snippet	Fix
3.c	Syntax	Missing semicolon	118	return 0	Add ;
3.c	Logical	Integer division	18	average = total / 5;	Use total / 5.0
3.c	Logical	Wrong comparison (min/max)	32	if(marks[i] > lowest)	Use < to find lowest
3.c	Array Bounds	Out of bounds access	42	for(i=0;i<=5;i++) attendance[i]	Change to i<5
3.c	Array Bounds	Out of bounds assignment	56	fees[5] = 100000;	fees size is 5
3.c	Array Bounds	Out of bounds access	96	for(i=0;i<=5;i++) inventory[i]	Change to i<5
3.c	Array Bounds	Out of bounds access	114	for(i=0;i<=5;i++) numbers[i]	Change to i<5
3.c	Buffer Overflow	strcpy overflow	48	strcpy(subject,"Programming")	subject[8] too small, need [12]
3.c	Logical	Assignment vs comparison	60	if(stock[2] = 30)	Use ==
3.c	Semantic	Array assignment	66	backup = stock;	Use memcpy or loop copy
3.c	Type Mismatch	Wrong format specifier	72	printf("%d", salary[0])	Use %f
3.c	Semantic	Missing & in scanf	80	scanf("%d", result[i]);	Use &result[i]
3.c	Semantic	Array assignment	86	city = "Jaipur";	Use strcpy
3.c	Semantic	Invalid array index	92	inventory["one"] = 100;	Use integer index
3.c	Semantic	Invalid array index	93	inventory[2.5] = 200;	Use integer index
3.c	Semantic	Array assignment	106	count = inventory;	Use memcpy or loop copy

File	Error Category	Error Type	Line	Code Snippet	Fix
3.c	Buffer Overflow	strcpy overflow	112	strcpy(course,"Computer")	course[6] too small, need [9]
3.c	String	Missing null terminator	100	college[0..4] set manually	Add college[5] = '\0'

Sheet: File 4 Errors

File	Error Category	Error Type	Line	Code Snippet	Fix
4.c	Syntax	Missing semicolon	122	printf("Report Generated Successfully")	Add ;
4.c	Logical	Integer division	14	averageSales = totalSales / 5;	Use totalSales / 5.0
4.c	Logical	Wrong comparison (min/max)	26	if(sales[i] < highestSale)	Use > to find highest
4.c	Array Bounds	Out of bounds assignment	36	monthlySales[5] = 250;	monthlySales size is 5
4.c	Array Bounds	Out of bounds access	38	for(i=0;i<=5;i++) monthlySales[i]	Change to i<5
4.c	Array Bounds	Out of bounds access	78	for(i=0;i<=5;i++) employees[i]	Change to i<5
4.c	Type Mismatch	Wrong format specifier	16	printf("Average Sales = %d", averageSales)	Use %f
4.c	Buffer Overflow	strcpy overflow	44	strcpy(product,"LaptopComputer")	product[8] too small, need [15]
4.c	Semantic	Array assignment	48	category = "Electronics";	Use strcpy
4.c	Semantic	Invalid array index	54	stock["first"] = 100;	Use integer index
4.c	Semantic	Invalid array index	55	stock[2.7] = 200;	Use integer index
4.c	Semantic	Missing & in scanf	64	scanf("%d", orders[i]);	Use &orders[i]
4.c	Logical	Assignment vs comparison	74	if(revenue[2] = 3000)	Use ==
4.c	Semantic	Array assignment	80	backupRevenue = revenue;	Use memcpy or loop copy
4.c	Type Mismatch	Wrong format specifier	86	printf("%d", commission[0])	Use %f
4.c	Semantic	Array assignment	90	city = "Jaipur";	Use strcpy
4.c	Buffer Overflow	strcpy overflow	106	strcpy(department,"Marketing")	department[6] too small, need [10]
4.c	Semantic	Array assignment	112	expenses = sales;	Use memcpy or loop copy
4.c	Array Bounds	Negative index	116	ratings[-1] = 5;	Use valid index 0-4
4.c	Type Mismatch	Wrong format specifier	124	printf("%s", feedback[0])	feedback is int[], use %d
4.c	String	Missing null terminator	94	branch[0..3] set manually	Add branch[4] = '\0'

Sheet: File 5 Errors

File	Error Category	Error Type	Line	Code Snippet	Fix
5.c	Logical	Integer division	18	average = total / 10;	Use total / 10.0
5.c	Logical	Wrong comparison (min/max)	28	if(marks[i] < highest)	Use > to find highest
5.c	Logical	Wrong comparison (min/max)	40	if(marks[i] > lowest)	Use < to find lowest
5.c	Array Bounds	Out of bounds assignment	52	attendance[10] = 0;	attendance size is 10 (indices 0-9)
5.c	Array Bounds	Out of bounds access	56	for(i=0;i<=10;i++) attendance[i]	Change to i<10
5.c	Array Bounds	Out of bounds access	130	for(i=0;i<=5;i++) stock[i]	Change to i<5

File	Error Category	Error Type	Line	Code Snippet	Fix
5.c	Array Bounds	Negative index	154	ratings[-1] = 10;	Use valid index 0-4
5.c	Array Bounds	Out of bounds access	170	printf temp[5]	temp size is 5, index 5 invalid
5.c	Type Mismatch	Wrong format specifier	20	printf("Average = %d", average)	Use %f
5.c	Buffer Overflow	strcpy overflow	64	strcpy(subject,"DataStructures")	subject[10] too small, need [15]
5.c	Semantic	Array assignment	68	department = "InformationTechnology";	Use strcpy
5.c	Semantic	Invalid array index	74	fees["two"] = 50000;	Use integer index
5.c	Semantic	Invalid array index	75	fees[4.5] = 70000;	Use integer index
5.c	Semantic	Missing & in scanf	84	scanf("%d", result[i]);	Use &result[i]
5.c	Logical	Assignment vs comparison	94	if(inventory[2] = 30)	Use ==
5.c	Semantic	Array assignment	100	backup = inventory;	Use memcpy or loop copy
5.c	Type Mismatch	Wrong format specifier	106	printf("%d", salary[0])	Use %f
5.c	Semantic	Array assignment	110	city = "Jaipur";	Use strcpy
5.c	Buffer Overflow	strcpy overflow	138	strcpy(course,"Programming")	course[8] too small, need [12]
5.c	Type Mismatch	Wrong format specifier	160	printf("%s", feedback[0])	feedback is int[], use %d
5.c	Semantic	Array assignment	166	scores = marks;	Use loop to copy first 5 elements
5.c	String	Missing null terminator	116	branch[0..3] set manually	Add branch[4] = '\0'